

Charlie Loitman, EIT

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EDUCATION

University of California, Berkeley, Master's of Climate Solutions August 2026

- Engineering and Data Science focus on power systems, renewable energy, grid operations, and electrification
- Conducted transmission expansion research using Gurobi, Python, and power flow modelling
- Worked with SunStrong Management using GridStatus data to predict optimal placement of residential battery systems
- Data Science for Energy, Energy and Environmental Markets, Electric Transmission Systems, Special Topics in Energy

Yale University – *cum laude* | *Tau Beta Pi* | *GRE: Verbal 168, Quant 170, Writing 6.0* December 2022

- ABET-Accredited B.S. in Mechanical Engineering *with distinction*; B.A. in Economics; Certificate in Energy Studies
- Engineering Capstone: A Method to Reduce Carbon Emissions from Concrete Usage in 3D Printing
- Economics Capstone: Hotelling vs Reality: Understanding the Dynamics of Global Oil Production

EXPERIENCE

Graduate Student Researcher | **Energy Modelling, Analysis, and Control Group (EMAC)**, Berkeley, CA April 2026 – Present

Disaggregating California energy demand for transmission expansion planning. Using sci-kit learn to project hourly substation level demand data through 2050 based on public and proprietary data. Applying Gurobi+Julia to evaluate transmission vs storage.

Graduate Intern | **Carbon Containment Lab**, New Haven, CT January 2026 – September 2026

Analyzed New Mexico oil well data using scikit-learn and PyTorch to identify abandoned oil wells and predict instances of methane leakage to prioritize plugging. Developing an economic royalty payment model to incentivize marginal wells to shut down.

Graduate Student Instructor | **UC Berkeley**, Berkeley, California July 2025 – May 2026

Wrote assignments and discussions for “Decision Analysis, Modeling, and Quantitative Methods” and “Marine Geochemistry.”

Trader | **Flow Traders**, Manhattan, New York February 2023 – March 2025

Automated and standardized futures-based ETFs; focused on commodities and the expansion into carbon credits, cryptocurrencies, and active energy strategies. Systematization of trading strategies and research reduced operational burden from six traders to two, doubled trading volumes, and increased profit 40% y-o-y. Co-created Centralized Execution Desk, developing strategies and a macro signal framework to improve trading across all products and optimizing order sizing and risk for each product and exchange. Became youngest member of global Trader Tech Team, coordinating product development and technology initiatives.

Summer Analyst | **Goldman Sachs, Customized Beta Strategies**, Manhattan, New York May – August 2022

Analyzed performance of risk models to allow tracking of custom benchmarks (S&P500, etc.) for sustainable investment goals (e.g. decarbonization). Created simulation tool used distributed to hundreds of financial advisers to demonstrate results across scenarios.

Research Analyst | **Ellington Management Group**, Stamford, CT January – August 2021

Conducted mathematical research to generate probability density functions for credit default swaptions to develop options strategy. Applied stochastic calculus and presented research in PhD seminar; developed a VIX trading strategy through options arbitrage.

Undergraduate Learning Assistant | **Yale University**, New Haven, CT August 2019 – December 2023

Proof-Based Linear Algebra; Proof-Based Vector Calculus; Thermodynamics, Fluid Dynamics; Dynamics; and Aerodynamics

Undergraduate Researcher | **MCubed Lab**, WashU, MO and **Qiu Lab**, Yale, CT May 2019 – December 2020

Applied ML classification with MCubed to identify novel stable high entropy alloys and double perovskites as candidates for solar panel materials. Won **George J. Schulz Fellowship in Physical Science** to fund DFT analysis of promising crystals for Qiu Lab.

AWARDS

4th Place | **UMich Ross Banking Design Dilemma**, Theme: Policy strategies for grid expansion February 2026

2nd Place Winner | **Rice Business Energy Case Competition**, Theme: Datacenter power demand November 2025

L.C. Lichty and E.O. Waters Prize in Mechanical Engineering | **Yale University** May 2023

“To the graduating senior who most demonstrates high scholarship and original research in mechanical engineering.”

Award for Outstanding Undergraduate Achievement | **Yale Science & Engineering Association** April 2023

Languages: Computer: Python, Julia, C++, SQL. Modelling: PyPSA, PyTorch, Gurobi, OpenDSS. Spoken: Spanish (intermediate)

Affiliations: Berkeley Energy and Resources Collaborative, Haas Climate Solutions Fund, Brooklyn Greenway Initiative